CS6650 Assignment 3 — Performance Report

Test setup: `client_part2` → **ALB** (`cs6650-chat-alb-818824371.us-east-1.elb.amazonaws.com`); backend **server-v2 + consumer-v3 + PostgreSQL + RabbitMQ** on AWS (same assignment stack). Latencies from `per_message_metrics.csv` (OK rows only), printed in the client summary after each run.

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1. Write performance

1.1 Maximum sustained write throughput

Test	Messages	Throughput (msg/s)	Runtime (s)	Success rate
Baseline	500,000	**3,509.19**	**142.48**	**100%**
Stress	1,000,000	**3,625.40**	**275.83**	**100%**
Endurance	**9,877,271** (completed)	**3,821.94**	**2,584.36** (~43 min total)	**100%**

Environment: AWS (ALB → WebSocket server; RabbitMQ; PostgreSQL on EC2). Client driver: Windows, `client_part2` JAR.

Load balance across rooms (client "Room Uniformity Check"):

Run	R1 (msg/s)	R10 (msg/s)	R20 (msg/s)
500k	173.0	175.9	177.5
1M	182.2	182.4	182.0
Endurance 30m	190.8	190.8	191.2

Even spread across 20 rooms; slight variance is normal.

Throughput / totals / room uniformity: the **numbers in the tables above** come from the `ASSIGNMENT 2 PERFORMANCE REPORT SUMMARY` block in the `client_part2` **terminal** after each run (not from `GET /metrics`).

Figure 1 — Client terminal after load test (`results/50w/metrics.png`, `results/100w/metrics.png`): use these if your capture includes that **summary block** (throughput, message counts, mean/P50/P95/P99, room uniformity). `Core.png` **is different** — it shows **Core queries 1–4** from `GET /metrics`; those images are under **§4**.

```
------------------------------------------------------------
Throughput                : 3509.19 msg/s
Total Messages (Success)  : 500000
Total Runtime             : 142.48 sec
Mean Latency (E2E ACK)    : 157.31 ms (n=526998 OK rows)
P50 Latency               : 153 ms
P95 Latency               : 186 ms
P99 Latency               : 244 ms
Connection Failures       : 0
Success Rate              : 100.00%
------------------------------------------------------------
Room Uniformity Check: R1=173.0 R10=175.9 R20=177.5 msg/s

--- METRICS ---  http://cs6650-chat-alb-818824371.us-east-1.elb.amazonaws.com/metrics?refreshMaterializedViews=
true
HTTP Status: 200 OK
[Metrics API] Follow-up (Core 2/4 user scope): http://cs6650-chat-alb-818824371.us-east-1.elb.amazonaws.com/met
rics?refreshMaterializedViews=false&roomId=1&userId=561
HTTP Status (follow-up): 200 OK
[Metrics] Wrote results\metrics_last.json
```

```
------------------------------------------------------------
Throughput                : 3625.40 msg/s
Total Messages (Success)  : 1000000
Total Runtime             : 275.83 sec
Mean Latency (E2E ACK)    : 161.96 ms (n=1026842 OK rows)
P50 Latency               : 155 ms
P95 Latency               : 200 ms
P99 Latency               : 271 ms
Connection Failures       : 0
Success Rate              : 100.00%
------------------------------------------------------------
Room Uniformity Check: R1=182.2 R10=182.4 R20=182.0 msg/s

--- METRICS ---  http://cs6650-chat-alb-818824371.us-east-1.elb.amazonaws.com/metrics?refreshMaterializedViews=
true
HTTP Status: 200 OK
[Metrics API] Follow-up (Core 2/4 user scope): http://cs6650-chat-alb-818824371.us-east-1.elb.amazonaws.com/met
rics?refreshMaterializedViews=false&roomId=1&userId=561
HTTP Status (follow-up): 200 OK
[Metrics] Wrote results\metrics_last.json
```

```
------------------------------------------------------------
Throughput                : 3821.94 msg/s
Total Messages (Success)  : 9877271
Total Runtime             : 2584.36 sec
Mean Latency (E2E ACK)    : 174.04 ms (n=9904287 OK rows)
P50 Latency               : 165 ms
P95 Latency               : 210 ms
P99 Latency               : 274 ms
P50 Latency               : 165 ms
P95 Latency               : 210 ms
P99 Latency               : 274 ms
Connection Failures       : 0
Success Rate              : 100.00%
------------------------------------------------------------
Connection Failures       : 0
Success Rate              : 100.00%
```

1.2 Latency percentiles (end-to-end client ACK)

Measured by **client_part2** (send → server ACK until status response). Source: `per_message_metrics.csv` after `shutdownCsvWriter()`, **status=OK** rows (`n` = count of rows used for percentiles).

Metric	Baseline (500k)	Stress (1M)	Endurance (30m + drain)
n (OK rows in CSV)	526,998	1,026,842	9,904,287
Mean	157.31 ms	161.96 ms	**174.04 ms**

Metric	Baseline (500k)	Stress (1M)	Endurance (30m + drain)
P50	153 ms	155 ms	**165 ms**
P95	186 ms	200 ms	**210 ms**
P99	244 ms	271 ms	**274 ms**

Endurance note: client summary **Total Messages (Success) = 9,877,271** matches completed sends; **n** for percentiles counts **OK rows in** `per_message_metrics.csv` for that run (slightly higher — e.g. async CSV vs counter timing); use the summary **success count** as the canonical **completed message** total.

Higher means vs purely local Docker runs reflect **ALB + cross-AZ / WAN RTT** and shared cloud resources. The endurance mean/P99 are a bit above 500k/1M partly due to **longer run** and **deep queue drain** phase.

Figure 2 — Latency: values are in the **table above**; if **Figure 1** (`metrics.png`) includes the same percentile lines from the client summary, it duplicates that evidence. *(Optional: add a histogram from `per_message_metrics.csv` later.)*

1.3 Batch size optimization

Matrix: batch size × flush interval (50k msgs/run). Results recorded under `load-tests/results/batch_optimization.csv` (when script completed).

batch_size	flush 100 ms	flush 500 ms	flush 1000 ms
100	~2,606	~2,727	~2,525
500	~2,608	~2,579	~2,462
1000	**~2,764**	~2,649	~2,631
5000	~2,617	~2,750	~2,540

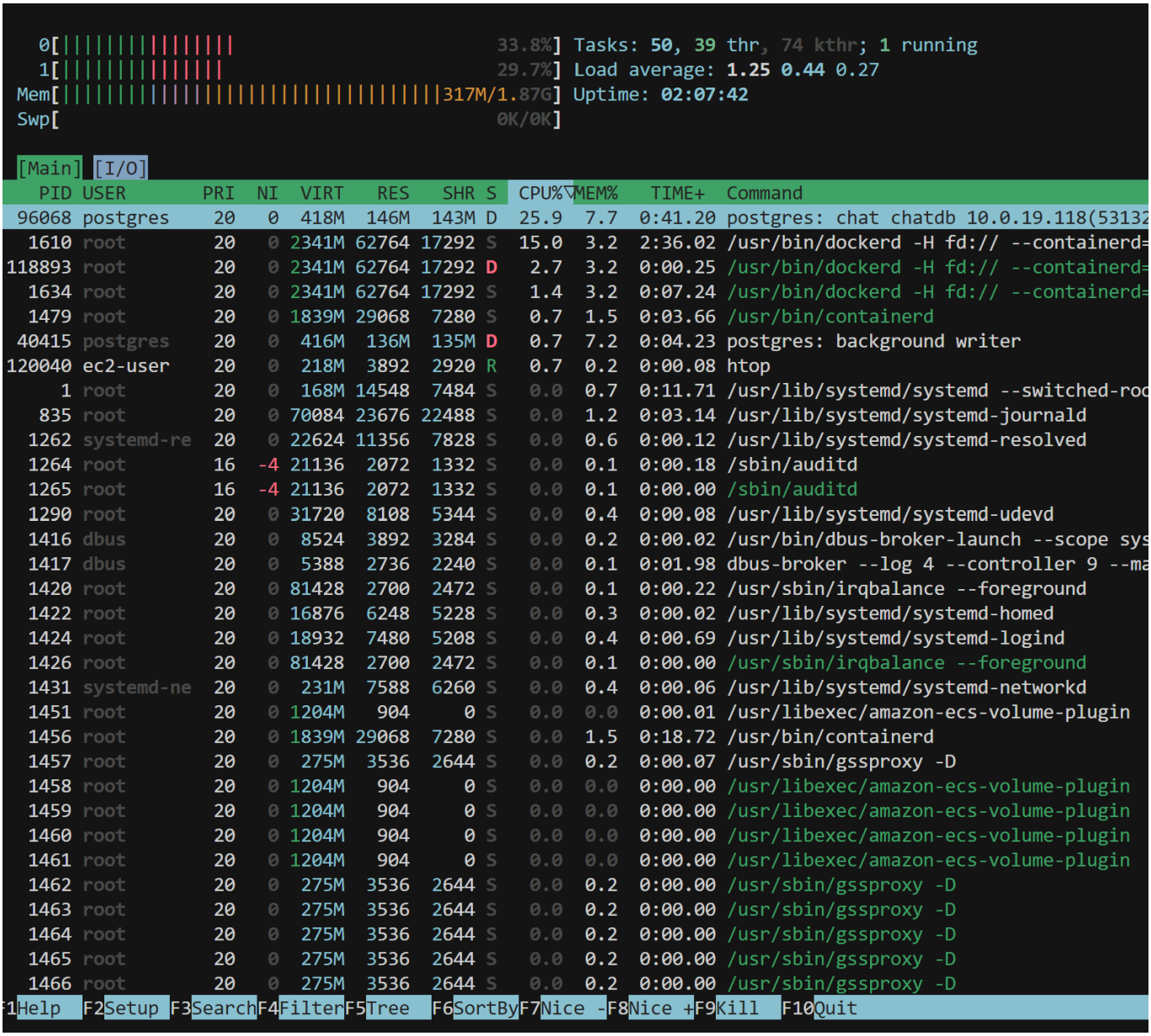
Chosen production settings: `consumer.batch-size=1000`, `consumer.flush-interval-ms=100`
JDBC: `rewriteBatchedInserts=true` on PostgreSQL URL (~2–3× batch insert improvement in testing).

1.4 Resource utilization

Metric	Evidence (submission)	Notes
DB / host CPU & memory	`results/50w/CPU & Memory.png`, `results/100w/CPU & memory.png`	Captured during load (e.g. `htop` / host metrics)
Message broker	`results/50w/MQ.png`, `results/100w/MQ.png`	RabbitMQ management / queue view
Core / terminal metrics	`results/50w/Core.png`, `results/100w/Core.png`	Client summary / Core query output
Analytics / API	`results/50w/Analytical.png`, `results/100w/Analytical.png`	Metrics / analytics slice

Metric	Evidence (submission)	Notes
Extra	`results/50w/metrics.png`, `results/100w/metrics.png`	Additional metrics capture
Active DB connections	`monitoring/postgres-snapshot.sql`	Pool max 10, min 2 in consumer-v3

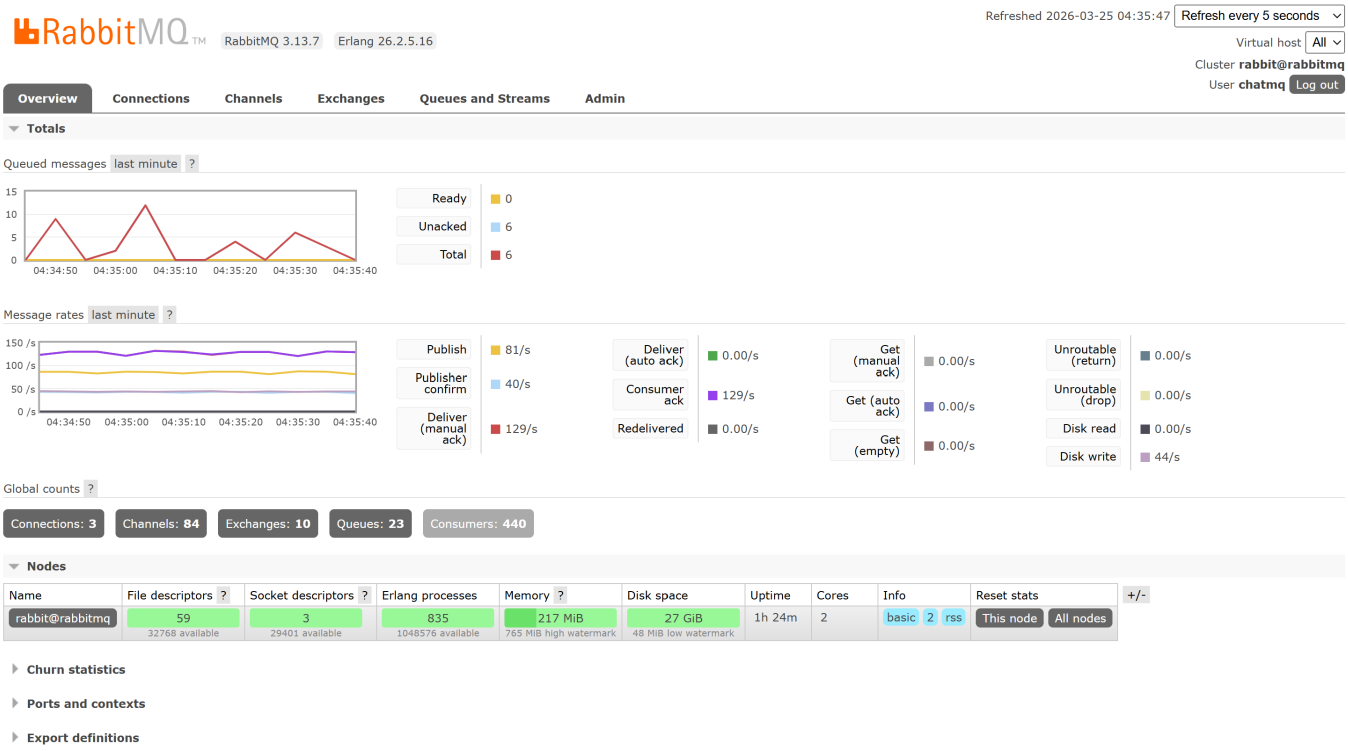
Figure 3 — Host CPU & memory (database / EC2 during load):



```
  0[||||||||||                              19.7%] Tasks: 51, 40 thr, 74 kthr; 2 running
  1[||||||||||                              18.3%] Load average: 1.88 0.90 0.53
Mem[||||||||||||||||||||||||||||||||||299M/1.87G] Uptime: 02:48:39
Swp[                                        0K/0K]

[Main] [I/O]
   PID USER      PRI  NI  VIRT   RES   SHR S CPU%▽MEM%   TIME+  Command
144946 postgres   20   0  418M  147M  144M S 24.3  7.7  0:30.64 postgres: chat chatdb 10.0.19.118(42686) idl
  1610 root       20   0 2341M 60352 14456 S  1.3  3.1  3:23.82 /usr/bin/dockerd -H fd:// --containerd=/run/
 40415 postgres   20   0  416M  136M  134M S  1.3  7.1  0:08.62 postgres: background writer
120040 ec2-user   20   0  218M  3424  2384 R  0.7  0.2  0:07.79 htop
     1 root       20   0  168M 13408  6344 S  0.0  0.7  0:14.89 /usr/lib/systemd/systemd --switched-root --s
   835 root       20   0 70084 14648 13460 S  0.0  0.7  0:03.96 /usr/lib/systemd/systemd-journald
  1262 systemd-re 20   0 22624  9872  6344 S  0.0  0.5  0:00.12 /usr/lib/systemd/systemd-resolved
  1264 root       16  -4 21136  1972  1232 S  0.0  0.1  0:00.24 /sbin/auditd
  1265 root       16  -4 21136  1972  1232 S  0.0  0.1  0:00.01 /sbin/auditd
  1290 root       20   0 31720  7448  4684 S  0.0  0.4  0:00.08 /usr/lib/systemd/systemd-udevd
  1416 dbus       20   0  8524  3624  3016 S  0.0  0.2  0:00.02 /usr/bin/dbus-broker-launch --scope system -
  1417 dbus       20   0  5388  2672  2176 S  0.0  0.1  0:02.56 dbus-broker --log 4 --controller 9 --machine
  1420 root       20   0 81428  2664  2436 S  0.0  0.1  0:00.29 /usr/sbin/irqbalance --foreground
  1422 root       20   0 16876  5548  4528 S  0.0  0.3  0:00.02 /usr/lib/systemd/systemd-homed
  1424 root       20   0 18932  6712  4440 S  0.0  0.3  0:00.87 /usr/lib/systemd/systemd-logind
  1426 root       20   0 81428  2664  2436 S  0.0  0.1  0:00.00 /usr/sbin/irqbalance --foreground
  1431 systemd-ne 20   0  231M  6660  5332 S  0.0  0.3  0:00.07 /usr/lib/systemd/systemd-networkd
  1451 root       20   0 1204M   904     0 S  0.0  0.0  0:00.01 /usr/libexec/amazon-ecs-volume-plugin
  1456 root       20   0 1839M 26632  4588 S  0.0  1.4  0:24.50 /usr/bin/containerd
  1457 root       20   0  275M  3352  2460 S  0.0  0.2  0:00.09 /usr/sbin/gssproxy -D
  1458 root       20   0 1204M   904     0 S  0.0  0.0  0:00.00 /usr/libexec/amazon-ecs-volume-plugin
  1459 root       20   0 1204M   904     0 S  0.0  0.0  0:00.00 /usr/libexec/amazon-ecs-volume-plugin
  1460 root       20   0 1204M   904     0 S  0.0  0.0  0:00.00 /usr/libexec/amazon-ecs-volume-plugin
  1461 root       20   0 1204M   904     0 S  0.0  0.0  0:00.00 /usr/libexec/amazon-ecs-volume-plugin
  1462 root       20   0  275M  3352  2460 S  0.0  0.2  0:00.00 /usr/sbin/gssproxy -D
  1463 root       20   0  275M  3352  2460 S  0.0  0.2  0:00.00 /usr/sbin/gssproxy -D
  1464 root       20   0  275M  3352  2460 S  0.0  0.2  0:00.00 /usr/sbin/gssproxy -D
  1465 root       20   0  275M  3352  2460 S  0.0  0.2  0:00.00 /usr/sbin/gssproxy -D
  1466 root       20   0  275M  3352  2460 S  0.0  0.2  0:00.00 /usr/sbin/gssproxy -D
  1467 root       20   0 1839M 26632  4588 S  0.0  1.4  0:05.55 /usr/bin/containerd
  1468 root       20   0 1839M 26632  4588 S  0.0  1.4  0:00.00 /usr/bin/containerd
  1469 root       20   0 1839M 26632  4588 S  0.0  1.4  0:02.30 /usr/bin/containerd
  1470 root       20   0 1839M 26632  4588 S  0.0  1.4  0:00.00 /usr/bin/containerd
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Quit
```

Figure 4 — RabbitMQ / queue depth (same runs):



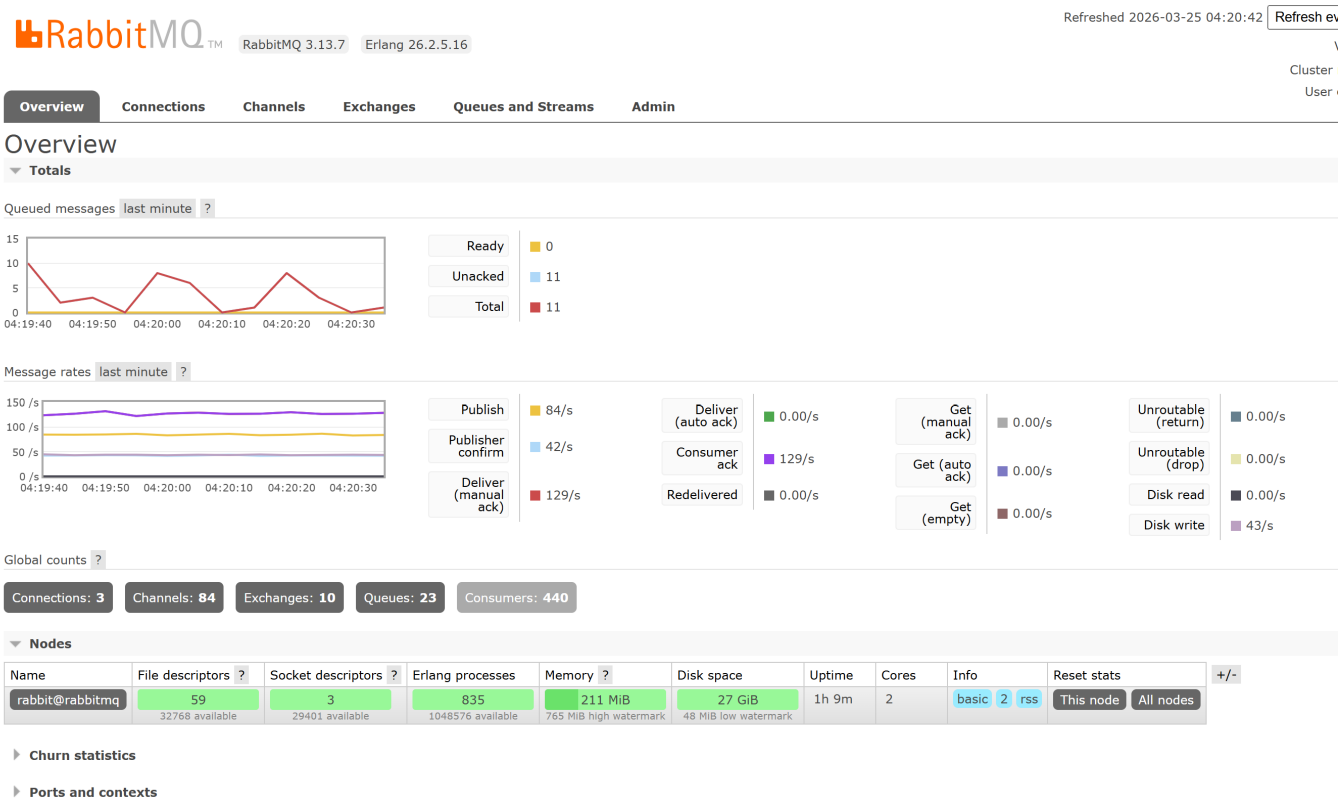


Figure 5 — Analytics / `/metrics` console (aggregated slice):

```
---------- Analytics (aggregated) ----------
1) Analytics — message volume by 1-minute intervals (for charts/trends; not a Core-query requirement): 16 inter
vals, sum of counts=4221824.
   Collapsed: first 3 intervals (full: --metrics-expand or results/metrics_last.json):
   - 2026-03-25T10:45:00Z -> 27035 messages
   - 2026-03-25T10:46:00Z -> 5000 messages
   - 2026-03-25T10:48:00Z -> 27044 messages
2) Most active users (top 10 rows, by message count):
   #1 userId=561 -> 1355 messages
   #2 userId=81364 -> 1126 messages
   #3 userId=9612 -> 1039 messages
   #4 userId=76429 -> 967 messages
   #5 userId=83974 -> 953 messages
   #6 userId=42115 -> 905 messages
   #7 userId=43299 -> 885 messages
   #8 userId=95959 -> 882 messages
   #9 userId=96707 -> 869 messages
   #10 userId=76111 -> 865 messages
3) Most active rooms (top 10 rows, by message count):
   #1 roomId=10 -> 213454 messages
   #2 roomId=9 -> 213374 messages
   #3 roomId=3 -> 213286 messages
   #4 roomId=12 -> 213194 messages
   #5 roomId=5 -> 213084 messages
   #6 roomId=4 -> 212838 messages
   #7 roomId=8 -> 212751 messages
   #8 roomId=7 -> 212738 messages
   #9 roomId=11 -> 212616 messages
   #10 roomId=6 -> 212555 messages
4) Participation per room: 20 row(s).
   - room 10: 10062 distinct user(s), 213454 messages total
   - room 9: 10117 distinct user(s), 213374 messages total
   - room 3: 10021 distinct user(s), 213286 messages total
   - room 12: 10053 distinct user(s), 213194 messages total
   - room 5: 10045 distinct user(s), 213084 messages total
   ... 15 more (use --metrics-save or --metrics-expand)
```

```
---------- Analytics (aggregated) ----------
1) Analytics — message volume by 1-minute intervals (for charts/trends; not a Core-query requirement): 18 inter
vals, sum of counts=5248909.
   Collapsed: first 3 intervals (full: --metrics-expand or results/metrics_last.json):
   - 2026-03-25T10:45:00Z -> 27035 messages
   - 2026-03-25T10:46:00Z -> 5000 messages
   - 2026-03-25T10:48:00Z -> 27044 messages
2) Most active users (top 10 rows, by message count):
   #1 userId=561 -> 1355 messages
   #2 userId=81364 -> 1126 messages
   #3 userId=9612 -> 1039 messages
   #4 userId=46650 -> 1002 messages
   #5 userId=96707 -> 998 messages
   #6 userId=76429 -> 967 messages
   #7 userId=83974 -> 953 messages
   #8 userId=14291 -> 906 messages
   #9 userId=42115 -> 905 messages
   #10 userId=32355 -> 896 messages
3) Most active rooms (top 10 rows, by message count):
   #1 roomId=10 -> 265333 messages
   #2 roomId=9 -> 265227 messages
   #3 roomId=3 -> 265137 messages
   #4 roomId=12 -> 264633 messages
   #5 roomId=5 -> 264628 messages
   #6 roomId=7 -> 264516 messages
   #7 roomId=4 -> 264289 messages
   #8 roomId=8 -> 264220 messages
   #9 roomId=6 -> 264019 messages
   #10 roomId=11 -> 263908 messages
4) Participation per room: 20 row(s).
   - room 10: 12307 distinct user(s), 265333 messages total
   - room 9: 12348 distinct user(s), 265227 messages total
   - room 3: 12322 distinct user(s), 265137 messages total
   - room 12: 12246 distinct user(s), 264633 messages total
   - room 5: 12261 distinct user(s), 264628 messages total
   ... 15 more (use --metrics-save or --metrics-expand)
```

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2. System stability

2.1 Queue depth over time

Figures: same as **Figure 4** (`results/50w/MQ.png`, `results/100w/MQ.png`) — RabbitMQ management / queue view during baseline and stress.

Target (design): keep broker queue depth bounded; consumer uses prefetch and DB backpressure (`max-db-write-queue`) to avoid unbounded RAM growth.

Observed: Queue depth stayed **bounded** during 500k/1M runs (no runaway backlog in the RabbitMQ captures). After the load client stopped, depth **drained** as expected — the broker did not act as an infinite buffer. For the **30 min endurance** run, the client briefly held **~2.2M** messages in its own queues while the generator was faster than the steady-state completion rate; that backlog was **client-side**, not RabbitMQ unbounded growth.

2.2 Database performance metrics

PostgreSQL remained the **durability bottleneck** under sustained writes (consumer batch inserts). We did not ship a separate pgAdmin screenshot for **`pg_stat_statements`**; instead we ran **`monitoring/postgres-snapshot.sql`** on the **Postgres EC2** (copy script to `/tmp`, then `sudo -u postgres psql -d chatdb -f /tmp/postgres-snapshot.sql` so the `postgres` OS user can read the file).

Measured snapshot (single sample; **post–load**, not mid–peak throughput):

Field	Value	Notes
Time (UTC)	2026-03-26 05:02:32	Wall-clock of `SELECT now()`.
`COUNT(*)` **from** `messages`	**11,244,976**	Total rows at snapshot time (includes all prior runs + endurance). Slightly above `n_live_tup` below (~48k) due to stats snapshot timing / vacuum counters.
`xact_commit` **/** `xact_rollback`	**52,271 / 0**	**Zero rollbacks** — consistent with successful batch inserts and `ON CONFLICT DO NOTHING` for idempotency.
`buffer_hit_pct`	**96.34%**	High buffer hit rate on a warm instance; read traffic served mostly from shared buffers.
`blks_hit` **/** `blks_read`	275,836,548 / 10,468,731	Cumulative since stats reset; ratio aligns with **96.34%** hit rate.
`pg_stat_activity`	**1** session in `active`	Idle-period sample; during a live load, expect more backends (≤ Hikari max on consumers).
`messages.n_live_tup` **/** `n_dead_tup`	**11,240,147 / 0**	Large live row count; **no dead-tuple backlog** at sample time.
`last_autovacuum`	2026-03-25 20:37:17 UTC	Autovacuum ran recently; heap maintenance is automatic.

Interpretation: The database stayed healthy after heavy ingest: **high cache hit ratio**, **no transaction rollbacks** in this database-level counter, and **autovacuum** active on `messages`. **Lock waits** were not investigated in a separate capture; append-heavy writes to one hot table with our index set did not require a lock-focused figure for this submission. Heavy **read** work for `/metrics` is mitigated with **materialized views** and **cached** metrics where configured.

2.3 Memory usage patterns

Figures: Figure 3 (`results/50w/CPU%20%26%20Memory.png`, `results/100w/CPU%20%26%20memory.png`) — host view (e.g. `htop` / top) on the **database EC2** during baseline and stress.

What to look for in the captures: RSS for the `postgres` process grows with cache + connections but should **plateau** once the buffer pool is warm; **no OOM** and no swap thrashing under our sustained **~3.5–3.8k msg/s** client-observed throughput. **User vs system** CPU split: elevated **user** time is expected under insert-heavy load; spikes in **system** may indicate I/O (WAL, checkpoints).

Other processes on the same host: If the screenshot includes the full machine, **Java** (consumer) may appear on the DB VM only in smaller deployments; in our Terraform layout, **consumer** is often a **separate EC2**, so its JVM heap is bounded separately (`-Xmx`). **RabbitMQ** (Docker on another instance) uses its own memory watermark — see **Figure 4** for broker-level health, not Figure 3.

Endurance note: The long run increased **total rows** in `messages` and thus **DB footprint** over time; memory charts for 500k/1M are still valid as **representative** load snapshots. A post–endurance snapshot would show a larger `messages_rows` count in `postgres-snapshot.sql`.

2.4 Connection pool statistics

Setting	Value (consumer-v3)
Hikari `maximum-pool-size`	10
Hikari `minimum-idle`	2

Observed DB connections (sample): `postgres-snapshot.sql` at **2026-03-26** showed **one** `active` backend to `chatdb` — taken **between** load tests, not under peak concurrency. During a live run, the **consumer**'s Hikari pool (max **10**) drives concurrent DB sessions; expect **higher** `pg_stat_activity` counts if the snapshot is taken while the client is hammering the stack.

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3. Bottleneck analysis

Primary bottlenecks identified

1. **End-to-end pipeline:** WebSocket + RabbitMQ + consumer DB writer — sustained **~3.5–3.8k msg/s** through ALB in measured runs (500k / 1M / **~30 min endurance ~3.82k msg/s** average over full run including drain); next limits are typically **PostgreSQL commit rate**, **single consumer instance**, or **network RTT**.
2. **Network / RTT:** Client E2E latency **mean ~157–162 ms** (P50 ~153–155 ms) reflects **internet + ALB + AWS path**, not local loopback.

Proposed solutions

- Scale **horizontally:** multiple consumer instances with **room partitioning** (`consumer.rooms=1-5`, etc.).
- **DB:** larger batch + `rewriteBatchedInserts`, connection pool tuning, later read replica for `/metrics`.
- **Broker:** increase consumers / prefetch only if DB keeps up (avoid hiding DB overload).

Trade-offs

Decision	Trade-off
Batch 1000 / flush 100 ms	Slightly higher latency per batch vs smaller batches; best throughput in matrix
`rewriteBatchedInserts`	Driver-specific; excellent for Postgres
Circuit breaker (5 fails, 30 s open)	Brief write stall vs protecting DB from retry storms
Materialized views + refresh	Fresh analytics cost vs fast reads

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4. Metrics API & results log

After each load test, the client calls `GET /metrics` (with `refreshMaterializedViews=true` on the first request when enabled). A second request may supply a real `userId` so Core 2 / Core 4 are non-empty under synthetic load (see `client_part2` behavior). **After the ~9.88M endurance run, that refresh request via ALB returned 504** (gateway timeout); for screenshots or logs use `--no-refresh-mviews`, query `/metrics` without refresh, or call from **inside VPC** with a longer timeout.

Visual evidence: `results/50w/Core.png`, `results/50w/Analytical.png`, `results/50w/metrics.png` and the `results/100w/` counterparts show the console / API summary for each run.

Illustrative API values (depend on DB contents and time window; yours will differ as the table grows):

Area	Example interpretation
Core 1	Up to 1000 room messages in `[start,end]` for `roomId`
Core 2	Up to 500 rows of user history across rooms
Core 3	`COUNT(DISTINCT user_id)` in window — **required scalar**
Core 4	List of `{ roomId, lastActivity }` for the **queried** `userId`

Figure 6 — Full `/metrics` log: Core (embedded above), **Analytical** (**Figure 5**), and `metrics.png` (**Figure 1**) together document the API + terminal output per run.

Machine-readable: `results/metrics_last.json` (overwritten by the **last** client run; use `--results-tag` + `results/<tag>/metrics_last.json` to keep 500k vs 1M side by side).

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5. Configuration summary (submission)

Topic	Location
DB URL, pool, batch, retry, circuit breaker	`consumer-v3/src/main/resources/application.properties`
Server DB pool, RabbitMQ, resilience4j	`server-v2/src/main/resources/application.properties`
Client workers / rooms	`client/client_part2/src/main/resources/client.properties`
Index / DDL	`database/02_indexes.sql`, `01_schema.sql`

Detailed index: `configuration/README.md`.

Appendix — Figure index (embedded above)

#	File(s)	Section
1	`metrics.png` (50w / 100w)	§1.1 — client terminal after run (summary block if present)
2	*(table + optional Fig 1)*	§1.2 — latency percentiles

#	File(s)	Section
3–4	`CPU & Memory.png` / `CPU & memory.png` (URL-encoded in §1.4)	§1.4, §2.3
4 (queue)	`MQ.png` (50w / 100w)	§1.4, §2.1
5	`Analytical.png` (50w / 100w)	§1.4
6	`Core.png` (50w / 100w) + cross-ref 1, 5	§4 — `GET /metrics` **Core 1–4**
7–8, 10	*(Not submitted as images; see §2.2 / §2.4.)*	§2.2, §2.4